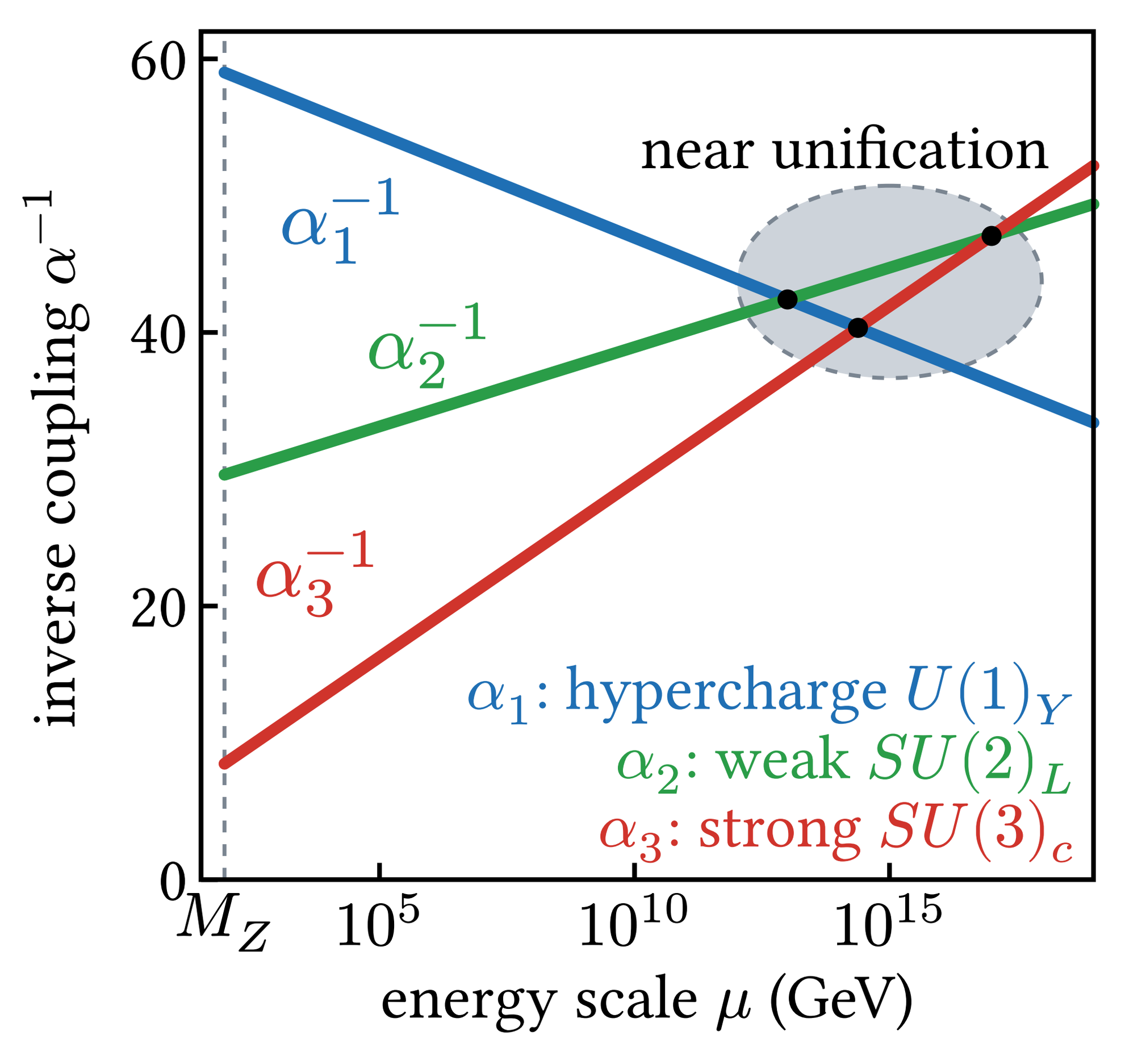
near unification
α_1^{-1}
α_2^{-1}
α_3^{-1}
α_1: hypercharge $U(1)_Y$
α_2: weak $SU(2)_L$
α_3: strong $SU(3)_c$
inverse coupling α^{-1}
energy scale μ (GeV)
0
20
40
60
M_Z
10^5
10^{10}
10^{15}